

New Discovery Could Explain What Gives Mars its Red Color

Scientists say they might have identified the mineral on Mars that is responsible for making the planet appear red.

New data collected on Mars through several methods suggests the substance, called ferrihydrite, is likely found in dust and rock material across wide areas of the planet. The team carrying out the research describes ferrihydrite as an “iron oxide mineral.”

The mineral appears reddish in color and can form in water-rich environments. The scientists said discovering large amounts of it on Mars adds to existing evidence that the planet might have held a large supply of liquid water in the past.

The research was carried out by an international team led by America’s Brown University and the University of Bern in Switzerland. The findings recently appeared in a study published in Nature Communications.

The researchers noted that scientists have long wondered about the exact substance that gives Mars – which is also known as the Red Planet – its deep red color. Past studies had suggested a mineral called hematite might be the cause.

The study is based on Martian data collected by several spacecraft. These include the Mars Reconnaissance Orbiter, operated by the American space agency NASA. In addition, information was captured by instruments aboard the European Space Agency’s (ESA’s) Mars Express and Trace Gas orbiters. Some data also came from exploring vehicles, called rovers, operating on Mars.

In their examinations, the researchers also used manmade, or synthetic, material designed to look and feel similar to real Martian dust.

Adomas Valantinas currently works as a researcher at Brown University’s Department of Earth, Environment & Planetary Sciences. He helped write the study. Valantinas noted the mystery of why Mars is red had been considered “for hundreds, if not thousands, of years.”

Valantinas recognized his team was not the first to consider ferrihydrite as a possible reason for the coloring of Mars. But he noted, “it has never been proven as we have now, using observational data and novel laboratory methods...”

Valantinas noted the discovery of widespread ferrihydrite suggests Mars once held liquid water that would have made it possible for the planet to support some form of life.

"Our study reveals that ferrihydrite formation on Mars required the presence of both oxygen... and water capable of reacting with iron," he added. "These conditions were very different from the dry, cold environment of Mars today."

John Mustard is a planetary expert at Brown University and was a co-writer of the study. He called it "a door-opening opportunity" for future research. But he added that additional study is needed to confirm the team's latest findings.

"As exciting as the new findings are, we realize that our results can only be verified by samples from Mars, which are currently being collected by NASA's Perseverance rover," Mustard said. He added, "When we get those back, we will be able to check whether our theory about ferrihydrite is correct."

I'm Bryan Lynn.

Bryan Lynn wrote this story for VOA Learning English, based on reports from

Vocabulary

novel - adj. ใหม่ หรือ แตกต่างจากสิ่งอื่นใด

opportunity - n. สถานการณ์ที่ทำให้ใครบางคนสามารถทำบางสิ่งได้

verify - v. ยืนยันว่าบางสิ่งเป็นจริงหรือไม่

check - v. ตรวจสอบว่าบางสิ่งถูกต้องหรือไม่

Reading Comprehension Quiz

1. What is the main subject of the scientific discovery mentioned in the article's opening paragraphs?

- The discovery of water on Mars
- A new type of Martian spacecraft
- The mineral responsible for Mars' red color
- Evidence of ancient life on Mars

2. According to new data, what is the name of the substance likely found in dust and rock material across wide areas of Mars, making it appear red?

- Hematite
- Limonite
- Ferrihydrite
- Magnetite

3. How do the researchers describe ferrihydrite in the study?

- A common silicate mineral
- An iron oxide mineral
- A carbon-based compound
- A hydrated sulfate

4. What specific environmental condition is mentioned as suitable for the formation of the reddish mineral ferrihydrite?

- Dry and cold climates
- Volcanic active regions
- Water-rich environments
- High atmospheric pressure zones

5. The discovery of large amounts of ferrihydrite on Mars supports which existing theory about the planet's past?

- Mars had a thicker atmosphere than Earth.
- Mars previously held a large supply of liquid water.
- Mars was once a gas giant planet.
- Mars was entirely covered in ice.

6. Which two institutions primarily led the international research team behind this study?

- NASA and the European Space Agency
- Brown University and the University of Bern
- Oxford University and MIT
- The Mars Reconnaissance Orbiter team and ESA

7. What mineral had previous studies commonly suggested as the cause of Mars' deep red color before this new research?

- Ferrihydrite
- Pyrite
- Hematite
- Goethite

8. All of the following spacecraft contributed Martian data to the study EXCEPT:

- Mars Reconnaissance Orbiter
- Mars Express
- Trace Gas Orbiter

9. In addition to data from Mars, what other type of material did the researchers use in their examinations?

- Lunar soil samples
- Synthetic material resembling Martian dust
- Earth-based red rocks
- Samples from meteorites

10. According to Adomas Valantinas, for approximately how long has the mystery of Mars' red color been a subject of consideration?

- Decades
- A few centuries
- Hundreds, if not thousands, of years
- Only since the space age began

11. What did Valantinas emphasize as novel about his team's work on ferrihydrite, compared to previous considerations?

- They were the first to consider ferrihydrite at all.
- They identified ferrihydrite's exact chemical composition.
- They proved its role using observational data and novel laboratory methods.
- They discovered ferrihydrite for the first time on Mars.

12. The discovery of widespread ferrihydrite on Mars implies what about the planet's past potential for life?

- Mars was always too cold to support life.
- Mars could have supported some form of life due to liquid water.
- Mars' atmosphere was toxic to all known life forms.
- Mars only had microscopic life existing in ice.

13. What two key elements were essential for ferrihydrite formation on Mars, according to the study's findings?

- Carbon dioxide and nitrogen
- Iron and volcanic activity
- Oxygen and water reacting with iron
- High temperatures and sulfur

14. How did John Mustard characterize the study's findings in terms of future research?

As a conclusive end to the mystery.

As a 'door-opening opportunity.'

As a minor contribution to existing knowledge.

As an area needing no further investigation.

15. According to John Mustard, what crucial step is needed to ultimately **verify the team's theory about ferrihydrite?**

More observations from orbiters.

Computer simulations of Martian conditions.

Analysis of rock samples collected by the Perseverance rover.

Conducting experiments on Earth using synthetic dust.